

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Installed, maintained, and troubleshoot advanced semiconductor equipment with complex components at customer sites, ensuring high-quality performance and adherence to safety protocols.
- Developed Python scripts to automate data extraction and transformation processes for operational reporting, reducing manual effort by approximately 15% per cycle.
- Managed production support for enterprise Java/.NET applications, responding to P1/P2 incidents and ensuring SLA adherence through proactive monitoring with **Splunk** and **Dynatrace**.
- Collaborated with experienced engineers on scheduled and predictive maintenance tasks, applying diagnostic techniques to analyze machine data and identify potential issues.
- Wrote and executed **SQL** queries for data validation and schema modifications across **PostgreSQL** and **Oracle** databases to support system integrations.
- Participated in structured, hands-on training programs focusing on semiconductor equipment applications, building deep technical skills in a highly specialized environment.
- Containerized applications using **Docker** for consistent deployment across development and production environments, integrating with **Jenkins** CI/CD pipelines.

SKILLS

Hardware & Systems – Semiconductor Equipment, Electrical Systems, Vacuum Systems, Mechanical Systems, Hydraulics, Pneumatics, Thermodynamics

Software & Tools – Python, Java, Spring Boot, SQL, PostgreSQL, Oracle, MySQL, Docker, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Methodologies – Agile, Scrum, CI/CD, Root Cause Analysis, Change Management, Predictive Maintenance, Continuous Improvement Programs (CIPs)

Data Analysis – Digital Diagnostics, Machine Data Analysis, Data Validation, Operational Reporting

General Engineering – Technical Schematics, Hand Tools, Digital Multimeters, Problem Solving, Zero-Defect Culture, Employee Safety

PROJECTS

Advanced Semiconductor Equipment Diagnostics and Maintenance

- Learned and applied structured procedures for scheduled and predictive maintenance on complex semiconductor manufacturing equipment, focusing on precision and quality.
- Utilized digital diagnostic tools and machine data analysis to troubleshoot and resolve technical challenges, contributing to a zero-defect culture.
- Supported customer operations by ensuring high equipment performance and adhering to strict safety and product quality standards in a cleanroom environment.

Automated Data Pipeline for Operational Insights

- Developed Python scripts to extract and process operational data from various sources, creating automated reports for system performance monitoring.
- Implemented data validation checks within the Python ETL process to ensure data integrity before generating reports, reducing data discrepancies by approximately 18%.
- Collaborated with operations teams to understand reporting requirements and refine the data pipeline for clearer, actionable insights into system health.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)